MA.7.AR.1.2

Benchmark

MA.7.AR.1.2 Determine whether two linear expressions are equivalent.

Example: Are the expressions $\frac{4}{3}(6 - x) - 3x$ and $8 - \frac{5}{3}x$ equivalent?

Benchmark Clarifications:
Clarification 1: Instruction includes using properties of operations accurately and efficiently.
Clarification 2: Instruction includes linear expressions in any form with rational coefficients.
Clarification 3: Refer to Properties of Operations, Equality and Inequality (Appendix D).

Connecting Benchmarks/Horizontal Alignment

- MA.7.NSO.1.2
- MA.7.NSO.2.1

Terms from the K-12 Glossary

Linear Expression

Vertical Alignment

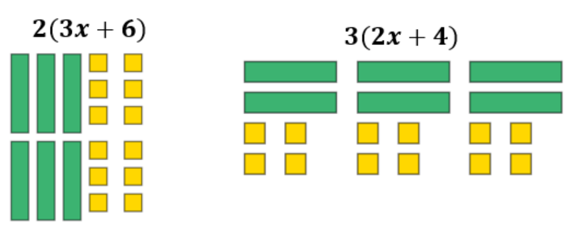
Previous Benchmarks
- MA.6.AR.1.4

Next Benchmarks
- MA.8.AR.1.1, MA.8.AR.1.3

Purpose and Instructional Strategies

In grade 6, students generated equivalent algebraic expressions with integer coefficients. In grade 7, students add and subtract linear expressions with rational coefficients as well as determine whether two linear expressions are equivalent. In grade 8, students will generate equivalent expressions including the use of integer exponents as well as rewrite the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions.

- Emphasize properties of operations to determine equivalence. Use manipulatives such as algebra tiles to emphasize the difference between linear and constant terms.
    - Algebra tiles can also be used to model operations concretely (*MTR.2.1*).



- Area Model can be used to represent the distributive property of multiplication over addition before moving to the abstract (*MTR.2.1*).

$$4(2x + 3) = 2(4x + 6)$$

From these representations, students can showcase equivalency between the two different linear expressions.

- Instruction includes students working within the same type of rational numbers when appropriate.
  - For example, if students are given fractions, the solution should be demonstrated in fractions and not be converted to decimals.
- Multiple equivalent expressions should be given, not just the most simplified. Transforming one or both expressions may be needed to show equivalence (*MTR.3.1*).
  - For example, $\frac{1}{2}x$ is equivalent to $\frac{1}{5}x + \frac{3}{10}x$ as well as $\frac{1}{10}x + \frac{1}{10}x + \frac{1}{10}x + \frac{1}{5}x$.

Common Misconceptions or Errors

- Students may incorrectly add and subtract terms that are unlike. To address this misconception, begin with a sorting activity to organize terms into groups before performing operations with those terms *(MTR.2.1)*. Have students verbalize the criteria for being considered like terms to ensure they are focused on the correct similarities.
- Students may incorrectly distribute the negative sign when subtracting a linear expression with more than one term.
  - For example, $\frac{1}{3} - (2x + \frac{7}{3})$ may be incorrectly rewritten as $\frac{1}{3} - 2x + \frac{7}{3}$ and concludes that it is equivalent to $\frac{8}{3} - 2x$.

Strategies to Support Tiered Instruction

- Teacher provides instruction to students that may incorrectly add and subtract unlike terms by giving students examples of like and unlike expressions and having students decide if they are equivalent. Give students the opportunity to explain why the expressions are equivalent or not equivalent.
- Teacher includes a leading coefficient of 1 in front of grouping symbols to help students recognize the implications of a negative sign in front of grouping symbols.
  - For example, $\frac{1}{3} - \left(2x + \frac{7}{3}\right)$ could be written as $\frac{1}{3} + (-1)\left(2x + \frac{7}{3}\right)$ or as $\frac{1}{3} + (-1) \cdot \left(2x + \frac{7}{3}\right)$ to help remind students to distribute the negative one before performing the next operation.
- Teacher provides instruction on using the properties of operations to group like terms together.
- Instruction includes the use of different color highlighters or shapes to identify algebraic terms and constants and then combining like terms identified with the same color.
  - For example, the expression $3x + 5 + \frac{2}{3}x - 6$ can be color-coded as $3x + 5 + \frac{2}{3}x - 6$ to determine that $3x + \frac{2}{3}x + 5 - 6 = 3\frac{2}{3}x - 1$.

- Teacher provides students with examples of linear expressions to explain whether two or more of the expressions are equivalent or not. Teacher co-creates examples of equivalent and non-equivalent linear expressions with students.
- Teacher co-creates a graphic organizer with students to review operations with rational numbers.
- Instruction includes providing students with two different linear expressions already represented with algebra tiles and then allowing the students to manipulate the algebra tiles to determine if they are equivalent. Students should be allowed to justify their reasoning for why the expressions are equivalent, or not.
- Teacher provides a sorting activity to organize terms into groups before performing operations with those terms. Have students verbalize the criteria for being considered like terms to ensure they are focused on the correct similarities.

Instructional Tasks

Instructional Task 1 (MTR.4.1)

Part A. Write three expressions that are equivalent to $\frac{4}{3}(6-x)+3\left(\frac{2}{3}x+1\right)$.

Part B. Compare those expressions with a partner. Think about the following questions to guide your conversation:

- How many are the same?
- How many are different?
- How many are correct?
- If there are incorrect expressions, what were the errors?

Instructional Items

Instructional Item 1

Match the following equivalent expressions.

	$3\frac{1}{8}x-1\frac{9}{10}$	$3\frac{1}{8}x-6\frac{3}{10}$	$3\frac{7}{8}x-1\frac{9}{10}$	$2\frac{3}{4}x-4$
$\left(3\frac{1}{2}x-4\frac{1}{10}\right)-\left(\frac{3}{8}x+2\frac{1}{5}\right)$				
$\left(3\frac{1}{2}x-4\frac{1}{10}\right)+\left(\frac{3}{8}x+2\frac{1}{5}\right)$				
$2\left(1\frac{1}{2}x-1\frac{3}{10}\right)+\frac{1}{2}\left(1\frac{2}{5}+\frac{1}{4}x\right)$				
$2\left(1\frac{1}{2}x-1\frac{3}{10}\right)-\left(1\frac{2}{5}+\frac{1}{4}x\right)$				

**The strategies, tasks and items included in the B1G-M are examples and should not be considered comprehensive.*